Chemical Kinetics

1. For the reaction $N_2 + 3H_2 \rightarrow 2NH_3$, how are the rate of reaction expressions inter-related $\frac{d[H_2]}{dt}$ and $\frac{d[NH_3]}{dt}$?

1) $-\frac{1}{3}\frac{d[H_2]}{dt} = +\frac{1}{2}\frac{d[NH_3]}{dt}$

2) $-\frac{1}{2}\frac{d[H_2]}{dt} = +\frac{1}{3}\frac{d[NH_3]}{dt}$

3) $+\frac{1}{2}\frac{d[H_2]}{dt} = -\frac{1}{2}\frac{d[NH_3]}{dt}$

4) $+\frac{1}{3}\frac{d[H_2]}{dt} = -\frac{1}{2}\frac{d[NH_3]}{dt}$

2. Consider the reaction: $2N_2O_4 \rightleftharpoons 4NO_2$

If $-\frac{d[N_2O_4]}{dt}$ = k and $\frac{d[NO_2]}{dt}$ = k′ then

1) 2k′ = k

2) k′ = 2k

3) k′ = k

4) k = $\frac{1}{4}$k′

3. For the reaction $2NH_3 \rightarrow N_2 + 3H_2$,

If $-\frac{d[NH_3]}{dt} = k_1[NH_3]$, $\frac{d[N_2]}{dt} = k_2[NH_3]$, $\frac{d[H_2]}{dt} = k_3[NH_3]$, then the relation between k_1, k_2 and k_3 is

1) $k_1 = k_2 = k_3$

2) $k_1 = 3k_2 = 2k_3$

3) $1.5k_1 = 3k_2 = k_3$

4) $2k_1 = k_2 = 3k_3$

4. Consider the reaction, $2N_2O_5 \rightarrow 4NO_2 + O_2$ In the reaction, NO_2 is being formed at the rate of 0.0125 mol L^{-1} s^{-1}. What is the rate of reaction at this time?

1) 0.0018 mol L^{-1} s^{-1}

2) 0.0031 mol L^{-1} s^{-1}

3) 0.0041 mol L^{-1} s^{-1}

4) 0.050 mol L^{-1} s^{-1}

5. For a reaction $R \rightarrow P$, the concentration of a reactant changes from 0.05 M to 0.04 M in 30 minutes. What will be the average rate of reaction in minutes?

1) $4 \times 10^{-4}\,\text{M min}^{-1}$

2) $8 \times 10^{-4}\,\text{M min}^{-1}$

3) $3.3\times10^{-4}\,\text{M min}^{-1}$

4) $2.2\times10^{-4}\,\text{M min}^{-1}$

6. In a reaction, $2HI \rightarrow H_2 + I_2$, the concentration of HI decreases from 0.5 mol L^{-1} to 0.4 mol L^{-1} in 10 minutes. What is the rate of reaction during this interval?

1) $5\times10^{-3}\,M\,\text{min}^{-1}$

2) $2.5\times10^{-3}\,M\,\text{min}^{-1}$

3) $5\times10^{-2}\,M\,\text{min}^{-1}$

4) $2.5\times10^{-2}\,M\,\text{min}^{-1}$

7. Nitrogen dioxide (NO_2) dissociates into nitric oxide (NO) and oxygen (O_2) as follows

$2NO_2 \rightarrow 2NO + O_2$

If the rate of decrease of concentration of NO_2 is $6.0\times10^{-12}\,\text{mol L}^{-1}\,\text{s}^{-1}$. What will be the rate of increase of concentration of O_2?

1) $3.0\times10^{-12}\,\text{mol L}^{-1}\,\text{s}^{-1}$

2) $6.0\times10^{-12}\,\text{mol L}^{-1}\,\text{s}^{-1}$

3) $1.0\times10^{-12}\,\text{mol L}^{-1}\,\text{s}^{-1}$

4) $1.5\times10^{-12}\,\text{mol L}^{-1}\,\text{s}^{-1}$

8. The rate of disappearance of SO_2 in the reaction, $2SO_2 + O_2 \rightarrow 2SO_3$ is $1.28\times10^{-5}\,\text{M s}^{-1}$. the rate of appearance of SO_3 is

1) $0.64\times10^{-5}\,\text{M s}^{-1}$

2) $0.32\times10^{-5}\,\text{M s}^{-1}$

3) $2.56\times10^{-5}\,\text{M s}^{-1}$

4) $1.28\times10^{-5}\,\text{M s}^{-1}$

9. In a reaction, $2X \rightarrow Y$, the concentration of X decreases from 0.50 M to 0.38 M in 10 min. What is the rate of reaction in M s^{-1} during this interval?

1) 1×10^{-4}

2) 4×10^{-2}

3) 2×10^{-2}

4) 1×10^{-2}

10. For the reaction, $2N_2O_5 \rightarrow 4NO_2 + O_2$ rate and rate constant are $1.02\times10^{-4}\,\text{mol L}^{-1}\,\text{s}^{-1}$ and $3.4\times10^{-5}\,\text{s}^{-1}$ respectively. The concentration of N_2O_5 in mol L^{-1} will be

1) 3.4×10^{-4}

2) 3.0

3) 5.2

4) 3.2×10^{-5}

11. The one which is a unimolecular reaction is

1) $2HI \rightarrow H_2 + I_2$

2) $N_2O_5 \rightarrow N_2O_4 + \frac{1}{2}O_2$

3) $H_2 + Cl_2 \rightarrow 2HCl$

4) $PCl_3 + Cl_2 \rightarrow PCl_5$

12. For a chemical reaction, C→D, it is found that the rate of reaction is doubled when the concentration of C is increased four times. The order of the reaction is

1) 1
2) 2
3) 1/2
4) zero

13. A chemical reaction involves two reacting species. The rate of reaction is directly proportional to the square of the concentration of one of them and inversely proportional to the concentration of the other. The order of the reaction is-

1) 1
2) 2
3) Zero
4) unpredictable

14. For the reaction $H_2(g) + Br_2(g) \rightarrow 2HBr(g)$, the experimental data suggests, $\text{Rate} = K[H_2][Br_2]^{1/2}$. the order for this reaction is

1) 2
2) $1\frac{1}{2}$
3) 1
4) $2\frac{1}{2}$

15. Calculate the order of the reaction w.r.t. A and B:

[A] (mol L^{-1})	[B] (mol L^{-1})	Rate (mol L^{-1} S^{-1})
0.05	0.05	1.2×10^{-3}
0.10	0.05	2.4×10^{-3}
0.05	0.10	1.2×10^{-3}

1) 1 and 0
2) 1 and 1
3) 0 and 1
4) None

16. For an elementary process 2P +Q → Z+ R, the molecularity is-

1) 2
2) 1
3) 3

4) Unpredictable

17. For a reaction A + B → products, the rate of the reaction was doubled when the concentration of A was doubled, the rate was again doubled when the concentration of both A and B were doubled the order of the reaction with respect to A and B are:

1) 1,1
2) 2,0
3) 1,0
4) 0,1

18. For the reaction A + B → products, it is found that the order of A is 1 and the order of B is $\frac{1}{2}$. When the concentration of both A and B are increased four times, the rate will increase by a factor of:

1) 16
2) 8
3) 6
4) 4

19. For a chemical reaction A → B, the rate of reaction doubles when the concentration of A is increased 8 times. The order of reaction w.r.t. A is:

1) 3
2) $\frac{1}{2}$
3) $\frac{1}{3}$
4) zero

20. For a gaseous reaction, rate = k [A] [B]. If the volume of the container is reduced to $\frac{1}{4}$ of initial, then the rate of the reaction will be _____ times of initial:

1) $\frac{1}{8}$
2) 8
3) $\frac{1}{16}$
4) 16

21. K for a zero-order reaction is $2\times10^{-2}\ \text{mol}\,\text{L}^{-1}\,\text{s}^{-1}$. If the concentration of the reactant after 25 s is 0.5 M, the initial concentration must have been.

1) 0.5 M
2) 1.25 M
3) 12.5 M
4) 1.0 M

22. The rate constant of a first-order reaction is $4\times10^{-3}\,\text{s}^{-1}$. At a reactant concentration of 0.02 M, the rate of reaction would be-

1) $8\times10^{-5}\,\text{M}\,\text{s}^{-1}$

2) $4\times10^{-3}\,Ms^{-1}$

3) $2\times10^{-1}\,Ms^{-1}$

4) $4\times10^{-1}\,Ms^{-1}$

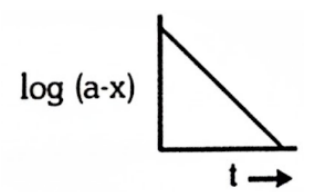
23. 99% of a first order reaction was complete in 32 min. when will 99.9% of the reaction complete?

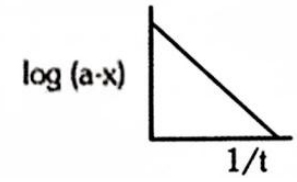
1) 50 min
2) 46 min
3) 49 min
4) 48 min

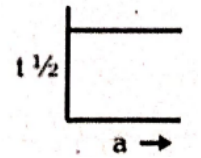
24. A first order reaction is carried out with an initial concentration of 10 mol per litre and 80% of the reactant changes into the product. Now if the same reaction is carried out with an initial concentration of 5 mol per litre for the same period the percentage of the reactant changing to the product is

1) 40
2) 80
3) 160
4) Cannot be calculated

25. Which of the following curves represents a 1st order reaction

1) 

2) 

3) 

4) 1 and 3 both

26. A reaction is found to have the rate constant x s^{-1}, by what factor the rate is increased if initial concentration of reactant is tripled

1) 3
2) 9
3) x
4) Remains same

27. Which is incorrect:

1) Half-life of a first order reaction is independent of initial concentration
2) Rate of reaction is constant for first order reaction
3) Unit of K for second order reaction is $mol^{-1}\,L\,s^{-1}$
4) Half-life of zero order is proportional to initial concentration

28. The expression which gives $\frac{1}{4}$th life of 1st order reaction is:

1) $\frac{K}{2.303}\log\frac{4}{3}$

2) $\frac{2.303}{K}\log 3$

3) $\frac{2.303}{K}\log\frac{3}{4}$

4) $\frac{2.303}{K}\log\frac{4}{3}$

29. The rate constant of a zero-order reaction is 0.2 mol dm-3 h^{-1}. If the concentration of the reactant after 30 minutes is 0.05 mol dm^{-3}. Then its initial concentration would be:

1) 6.05 mol dm^{-3}

2) 0.15 mol dm^{-3}

3) 0.25 mol dm^{-3}

4) 4.00 mol dm^{-3}

30. Correct statement about first order reaction is:

1) $t_{completion} = \text{finite}$

2) $t_{1/2} \propto \frac{1}{a}$

3) Unit of K is mol $\text{lit}^{-1}\,\text{s}^{-1}$

4) $t_{1/2} \times K =$ constant (at given temperature)

Answer Key

1	1	**16**	3
2	2	**17**	3
3	3	**18**	2
4	2	**19**	3
5	3	**20**	4
6	1	**21**	4
7	1	**22**	1
8	4	**23**	4
9	1	**24**	2
10	2	**25**	4
11	2	**26**	1
12	3	**27**	2
13	1	**28**	4
14	2	**29**	2
15	1	**30**	4